Lesson Plan · Wednesday, November 11

Algebra I · Mr. Pandey · Module 3, Topic 1, Lesson 3: Modeling Linear Inequalities (Day 1 of 3) · essentials only (✪)

Objectives (students will…)

- Write and analyze a linear function from a context (rate of change, y-intercept).
- Use a graph to solve inequalities.
- Graph solution sets on a number line with open or closed circles.

TEKS

- **A.2C** write linear equations from a context
- **A.5B** solve linear inequalities in one variable
- Process: A.1A, A.1D

Key terms

- linear inequality

Materials

- Student Edition pp. 371–375 (Getting Started, Activity 3.1)
- Inequality-words chart (at least ≥, at most ≤, more than >, fewer than <)
- Rulers for the horizontal-line method
- Whiteboards

Bell work (on the board)

- "at least" means which symbol? ≥
- "no more than" means which symbol? ≤
- "fewer than" means which symbol? <
- Solve 3.75b + 25 = 400 b = 100

Agenda (60 minutes)

Time	Activity	What to do / say
0–5	Bell work	Build the inequality-words chart on the board from Q1–3 (TE EB tip). Leave it up for all of Lesson 3.
5–20	✪ Getting Started: Fundraising Function · SE p. 372	Elijah: $25 credit + $3.75 per box. Q1 f(b) = 3.75b + 25. Q2 independent boxes sold, dependent total sales ($); slope 3.75, dollars per box; y-intercept 25, the starting credit.
20–45	✪ Activity 3.1: Modeling Linear Inequalities · SE pp. 373–375	Q1 goal $1600: horizontal line → 420 boxes. Define *linear inequality*. Q2 below the line: f(b) ≤ 1600, b ≤ 420, closed circle; only whole numbers of boxes make sense. Q3 above: b > 420, open circle. Q4 open = not included, closed = included. Q5 at least $925 b ≥ 240; less than $2050 b < 540; exactly $700 b = 180.
45–55	Exit ticket	f(b) = 3.75b + 25. Write an inequality and graph it on a number line. 1. sales of at most $1150 b ≤ 300, closed circle, shade left 2. sales of more than $475 b > 120, open circle, shade right IEP/504: read aloud on request, properties/notes card, extra time.
55–60	Wrap up	Homework: Lesson 3 Assignment Practice 1a–d (SE p. 383). Tomorrow: solve inequalities with algebra.

Check for understanding

- Whiteboards: "Open or closed for x < 3?" open
- Point to the graph: "Where is f(b) > 1600?" above the line
- Activity 3.1 Q1–5 are the TE's check points.

After class

- Missed circles/shading → Skills Practice Set B (number lines).
- Missed writing the inequality → Skills Practice Set A.
- Re-teach the words chart at tomorrow's bell work.

Getting Started and 3.1 are starred. TE pacing is 3 days (11/11, 11/12, 11/16), matching the district calendar.